

[RFC] RMS menu retranslation after auto translation is cleared

High-level design

What is the problem to be solved?

What is the solution and why it can be solved?

Detail Design / Checklist guideline [Optional]

Event Schema Changes

RMS → i18n

i18n consume event flow

Architecture design/changes

SLOs

Service dependencies & compatibility

3rdParty Integration

Availability, Uptime, Disaster Recovery planning

Performance consideration/testing

Troubleshooting/Logging Considerations

Security considerations

Am I implementing business logic, permission checking, authentication, security controls on the server side, not on the client-side?

Please list all PII, sensitive information, credentials, tokens, etc. that are introduced by this feature, where are those stored, and what are the purposes.

Please list integration with external parties and security controls to be implemented

Please list endpoints that are introduced or modified by this feature. Describe authentication and authorization are implemented.

Security checklist

Testing & rollout

Metrics & Monitoring

Customer support considerations

Read the guide about RFC: [📖 Request for Comments \(RFC\) Guide](#)

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Security approval	TO REVIEW
Review complete date	

High-level design

Link to jira: [🔍 PIN-10845: Menu retranslation after auto translation is cleared](#) OPEN

What is the problem to be solved?

When either the source text or the Thai text of an RMS menu entry is modified, the existing auto-translation in RMS is cleared. However, on the i18n side, the message event may not trigger a new translation if the `translation key`, which is calculated from the source text, does not change. As a result, the i18n service does not perform a new translation or push an updated auto-translation back to RMS, leaving the RMS auto-translation field empty after edits.

What is the solution and why it can be solved?

Instead of relying solely on the `translation key` to decide whether to translate and sync results, the i18n service will use the presence of translation content. If any language's content is empty, i18n will trigger auto-translation and push the result back to RMS. To enable this, the menu-changed event will include the per-language translation content. Based on this payload, i18n will determine which languages are missing and backfill them automatically, ensuring gaps are corrected regardless of why they occurred.

- Event payload change: include per-language translation content in the menu change event.
- i18n decision logic change: trigger translation based on empty/missing content, not only translation key changes.
- Sync missing translations only: generate/refresh translations for missing languages and sync results back to RMS.

Detail Design / Checklist guideline [Optional]

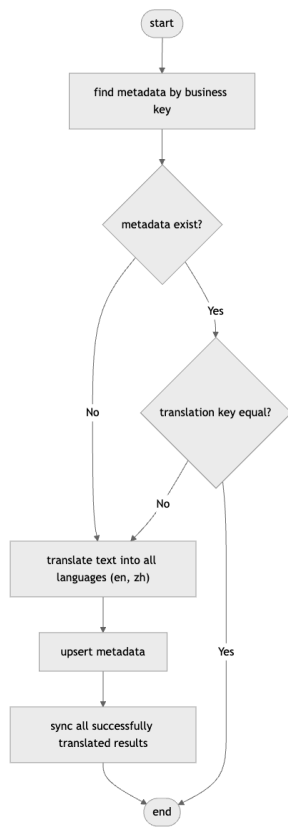
Event Schema Changes

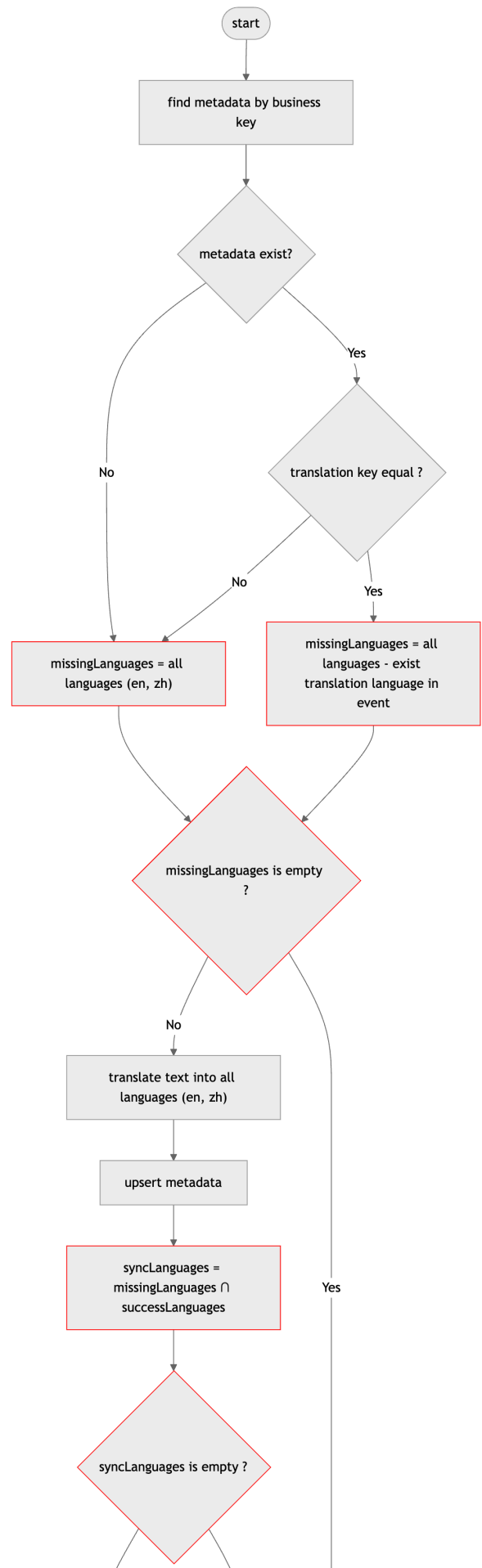
RMS → i18n

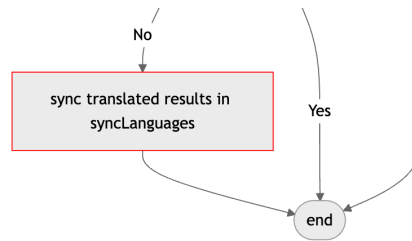
name	type	description
Id	string	generic id e.g. restaurant id
SubId	string	sub item id

		e.g. menu_id, menu_item_property_id
SourceLanguage	enum	• en • th
Text	string	text in Thai and need to translate
MultiLanguageText (New added)	map[string]string	<pre>{ "th": "ข้าวผัดกุ้ง", "en": "Fried rice with shrimp", "zh": "虾仁炒饭" }</pre>
Service	enum	Service name • menu
Category	string	• menu_item_name • menu_item_description • menu_group • menu_item_property • menu_item_property_value
Option	TranslateOption	set custom translate option

i18n consume event flow







Architecture design/changes

Think and write about these points

- Are you deploy a new service?
 - No
- Is there any change in request flow?
 - Yes
- Do you move data to a new table?
 - No
- Is your system has a new dependency?
 - No
- Do you store new data? how large? where?
 - No

SLOs

Endpoint	Current req/min	Expected req/min	Current p99 response time	Expected p99 response time	Reason of impact

Service dependencies & compatibility

- Who is consuming/will consume this thing?
 - RMS
- Who do I need to make a request?
- How my service will be operate if my dependant is unavailable?

- Menus with failed translations will be added to the pending list.
- Do you have an async call? pub/sub? MQ? who will use my data?
 - RMS → Im-i18n
 - When menu is changed, change event should be send to i18n
 - i18n → RMS
 - When menu is translated, translated result should be send to RMS
- If I consume message from MQ, what will happen if there is error while processing message? Discard? Retry? Dead letter? Alert?
 - Retry
- Does it break API backward compatibility? How long you will support the older API version?
 - No
- Does the service has new enum type/add new enum value/use new integer/string value as a categorical type?
 - Make sure that you implement feature flag to separate deployment and release
 - Make sure to handle (and always handle) unknown value accordingly
- **[LINEMAN]** Is there any data in this feature that must be removed after clients delete their accounts (e.g. credit card data) or be transferred to the new account (e.g. fraud, ban, or order stat from the previous account)
 - No
- **[LINEMAN]** Is the change related to the Payment area? If yes, please recheck the [Checklist for Payment Related Changes](#)
 - No

3rdParty Integration

If this RFC intends to integrate with 3rd Party Service, please follow this guideline [02 - Guideline - 3rdParty Integration Guideline](#)

Checklist Topic	Answer
Integration Document	
Authentication method	

File Sharing	
Encryption on file	
Circuit Breaker	
Fallback/Backup	
Monitoring & Alerting	
Escalation Process	
Cost & Billing	

Availability, Uptime, Disaster Recovery planning

Fill in the table below that is related to this RFC.

New Dependencies	Purpose of usage	Recovery plan / degraded behavior	Manual testing & automate testing method for DR

Note: For the recovery plan, please make sure once the plan is executed, the system itself should remain functioning correctly.

Also, checking for these common mistakes

- If this service or this RFC changes are heavy workload or has a high usage growth
 - Separate physical DB instances
 - Separate physical MQ instances
 - Separate physical Cache/Redis instances
 - Implement data partitioning

- Implement sharding according to the workload
- Implement a data retention policy
- Implement data archival policy by moving old data into a separate tier
- Make sure all existing dependencies are recoverable or gracefully degraded without/with minimum impact on functionality
- Separate mission-critical RPC endpoints from non-critical RPC endpoints by deploying into a separate set of machines
- [Verda] Prefer Physical Machine over VM for a heavy workload
- DR, performance test, and fault-tolerant requirements should be automate tested before the production release.

Performance consideration/testing

- Does it increase number of DB queries or change query?
 - How many will it increase? in what BigO? eg. $O(1)$, $O(n)$, more? and what's n ?
 - Have you take a look that it doesn't has $1+N$ query patterns?
 - Does it hit DB index?
 - Is it fast to query? Usually, should be less than 20ms when you try it on phpMyAdmin or DB query tools
 - Does it return small number of rows? Usually, should be less than 500 rows per query
- Does it increase number of DB insert/update/delete?
 - How many of rows will be updated per single transaction? Usually, should be less than 10.
 - Can you break it into smaller transaction if the number of rows are over 10?
 - How long a single transaction will be held? Usually, should be less than 100ms.
 - Can you use Redis distributed lock instead if it hold a long transaction?
- Does it increase numbers of network calls (eg. send HTTP request, gRPC request, publish to MQ)
 - How many will it increase? in what BigO? eg. $O(1)$, $O(n)$, more? and what's n ?
 - Can you make the network calls being process in parallel? all at the same time?
 - Can you batch many requests into a single request if it cannot send in parallel?
 - Is it small on the request size? Usually, should be less than 100KB.
 - Is it small on the response size? Usually, should be less than 1MB (uncompressed).
 - If it is large for request or response size, please consider to replicate data to your service instead.

Troubleshooting/Logging Considerations

1. Where can you investigate the issue, and what key information needs to be logged for effective troubleshooting?

Consider what critical fields (e.g., rid, orderId, userId, timestamps, error messages) need to be captured in logs, and identify where (e.g., application logs, server logs, monitoring tools) you can investigate when issues arise.

ANSWER:

2. What is the starting point for troubleshooting , and how can you ensure that logs contain enough context to trace the root cause?

Define the primary location for initial investigation (e.g., microservices logs, network logs, database logs) and determine if the logs contain sufficient contextual information, such as trace IDs or correlation IDs, to identify the source of the issue.

ANSWER:

Security considerations

For full checklist, please consider create & link to another wiki using template [RFC Security Checklist for Code changes](#) .

Am I implementing business logic, permission checking, authentication, security controls on the server side, not on the client-side?

Yes, those logics are implemented on the server side / No, those logics are implemented on the client side because of ...

Please list all PII, sensitive information, credentials, tokens, etc. that are introduced by this feature, where are those stored, and what are the purposes.

Data	Storage	Purpose	Encryption

Please list integration with external parties and security controls to be implemented

Name of external party	Use cases	Security controls

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Please list endpoints that are introduced or modified by this feature. Describe authentication and authorization are implemented.

Endpoint	Authentication	Authorization + Permission Roles

Security checklist

- Make sure the system denies access by default if the request is not authenticated.
- Make sure that all error messages don't contain sensitive info – stack trace, class name, software name, or library name.
- Make sure user inputs are validated before processing the request – according to business logic, fraud, malicious user.
- Make sure user inputs are sanitized before processing the request – do not contain JS, HTML, code, invisible characters, character control, binary data, etc.
- Make sure all outputs are sanitized and escaped – do not contain JS, HTML, etc.
- Make sure all application logs (**including server & web browser & mobile app**) don't contain any sensitive info, credential, PII data, etc.
- Make sure firewall and network ACL are properly configured.

Testing & rollout

- How to check whether this thing is working properly? (Especially in Production env)
 - Create a menu on RMS and ensure it can be translated normally
 - Modify Thai name of the menu, it can be re-translated normally
- How to disable this feature without rebuild/redeploy this service if something went wrong
 - feature flag
 - Disabled translation sync flag for all languages
- Is it safe to rollback service to old code?
 - Yes
- What is a new automated regression test case and what will break?

- Did you put logs to help records what happened and debugging on production?
 - ~~Audit log — who did it, what is the action?~~
 - Event log - eg. at the end of controller, at the important point?
 - Yes
- Steps to rollout
 - ~~Database Schema Migration~~
 - ~~System configuration changes~~
 - ~~Network configuration/Network ACL~~
 - Service Deployment
 - menutransl8
 - i18n-service-consumer
 - ~~Seed data~~
 - ~~Metrics to check~~
 - ~~Feature flag configuration~~
 - ~~Smoke test~~

Metrics & Monitoring

Metric Name (Biz & Sys)	Graph Type	Condition	Dashboard Name

Customer support considerations

- ~~What CS needs to know about this task?~~
- What date this feature will be released?
 - 2026年6月3日
- ~~Where is admin tools for CS to check for data about this task?~~